COMPUTING IRREDUCIBLE DECOMPOSITION OF MONOMIAL IDEALS

SHUHONG GAO, MINGFU ZHU, AND SHUAI WEI

Abstract. We give two algorithms for finding the irreducible decomposition of monomial ideals. The first one is based on the projection and the staircase structure of monomial ideals. The second one is based on a simple enumeration algorithm to traverse all the facets of the Scarf complex associated with a monomial ideal.

1. Abstract

2. Introduction

3. Monomial Ideals

4. Tree representation and operations

Note that monomials are represented by vectors in $\mathbb{N}^n$ and irreducible components are represented by vectors in $\overline{\mathbb{N}}^n$. To efficiently represent a collect of vectors, we use a tree structure. This is used in **??**. This data structure is also widely used in computer science, where it is called a trie.

Tree representation. Suppose $S \subseteq \mathbb{N}^n$ is a set of vectors corresponding to the generators of a monomial ideal $I \subseteq \mathbb{K}[X]$. We represent S as a rooted tree $\mathscr{T}$ if height n in a natural way. The tree should have $|S|$ leaves and the unique path of the tree from the root to a leaf represents a vector in S. Precisely, to represent a vector $\alpha = (a_1, \ldots, a_n)$, we label all the notes except the root of the path simply by $a_1, \ldots, a_n$ in the order from the root to the leaf. We regard the root as being at height 0. For two vectors $\alpha = (a_1, \ldots, a_n)$ and $\beta = (b_1, \ldots, b_n)$, if $a_j = b_j$ for $1 \leq j \leq k$ but $a_{k+1} \neq b_{k+1}$, then α and β share their corresponding path until hight k. Furthermore, the children of each node in the tree are listed in increasing order with respect to their coordinates. Figure 1 is the tree representation for $I = \langle z^4, y^4, x^2y^2z^3, x^2y^3z, x^3yz^2 \rangle$.

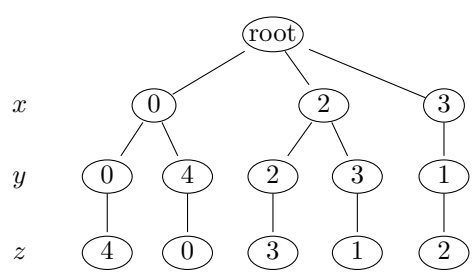

Figure 1. An example of tree representaton.

The tree representation for a set of irreducible components could be constructed in a similar manner.

To perform the operations on sets of vectors, we need only perform on trees. We need three basic tree operations: MaxMerge and MinMerge.

MaxMerge. Let A and B be two sets of vectors such that any element γ from A or B, we have $\gamma \in \mathbb{M}^n$, where $\mathbb{M} = \mathbb{N} \sqcup \{\infty\}$. We use $\mathrm{MaxMerge}(A, B)$ to represent the set of maximal elements in $A \cup B$. For two vectors α, β in $A \cup B$, if $\alpha \leq \beta$, then α should be removed in this operation. Let I_1 be the ideal generated by $\{m^\alpha \mid \alpha \in A\}$ and I_2 the ideal generated by $\{m^\beta \mid \beta \in B\}$. Then $\{m^\gamma \mid \gamma \in \mathrm{MaxMerge}(A, B)\}$ represents the set of irreducible componenets of the ideal $I_1 \cap I_2$.

MinMerge. Let $\mathscr{T}_1$ and $\mathscr{T}_2$ be two rooted tree with the same height. Merge $\mathscr{T}_1$ and $\mathscr{T}_2$ to form a new rooted tree $\mathscr{T}$ with the same height such that repetition is ignored. For two paths $\wp_1, \wp_2$ in $\mathscr{T}$, letting α, β be the corresponding vector, respectively, if $\alpha \leq \beta$, i.e., $x^\alpha \mid x^\beta$, then the path $\wp_2$ should be removed in this operation. The purpose is to find the minimal generating set for the ideal $I_1 + I_2$, where $\mathscr{T}_1, \mathscr{T}_2$ is the tree representation for I_1, I_2, respectively.

5. Structure properties of monomial bases

Theorem 5.1. $\mathrm{Irr}(I) = \bigsqcup_{k=1}^{s} \big(\mathrm{Irr}(I_{k-1}) \smallsetminus \mathrm{Irr}(I_k)\big) \otimes d_k$, *which is a disjoint union.*

Recursive Algorithm: $\mathrm{Irr}(\mathscr{T}, l)$.
Input: $\mathscr{T}$, a nonempty tree encoding $\mathrm{Min}(I)$ and the number of levels (variables) l
Output a set representing $\mathrm{Irr}(I)$

Step 1. Start at the root of $\mathscr{T}$. If the height of $\mathscr{T}$ is 1, then $\mathscr{T}$ only have 1 child (leave), let d be the label of that child and let $S := [[d]]$, where S works like a matrix or a 2D array. Return S.

Step 2. Now $\mathscr{T}$ has height at least two. Suppose $(0 \leq)\, d_0 < d_1 < \cdots < d_{s-1}$ are the labels of the children under the root of $\mathscr{T}$, and let $\mathscr{T}_k$ be the subtree extending from d_k, $0 \leq k \leq s-1$. Note that the root of $\mathscr{T}_k$ is the node labeled by d_k, but now unlabeled.

Step 3. Set

$$S := \begin{cases} [[\underbrace{\infty, \ldots, \infty}_{l-1}, d_0]], & \text{if } d_0 > 0, \\ [\ ] \text{ or } [[\ ]], & \text{otherwise.} \end{cases}$$

If $d_0 > 0$ here, we think $d_{-1} = 0$.

Step 4. Assume x_k is the variable corresponding to the children of $\mathscr{T}$. Set

$$d_s = \begin{cases} d_{s-1}, & \text{if } I \text{ contains the power of } x_k, \\ \infty, & \text{otherwise.} \end{cases}$$

Note that if I contains the power of x_k, say x_k^e, then e must be the largest label on the labels of childrens of $\mathscr{T}$, i.e., the label of the right-most child $\mathscr{T}_{s-1}$ of $\mathscr{T}$. So we only need to check whether the labels from the first child of $\mathscr{T}_{s-1}$ to the left-most leaf are all 0 or not.

Step 5. For k from 0 to $s-1$ do:
$S := \mathrm{MaxMerge}\big(S, \mathrm{Irr}(\mathscr{T}_k, l-1) \otimes d_{k+1}\big)$.
$\mathscr{T}_{k+1} := \mathrm{MinMerge}(\mathscr{T}_k, \mathscr{T}_{k+1})$.

Step 6. Return S.

Note that when I contains the power of x_k, for $k = s-2$ in step 5, $\mathscr{T}_{s-1} = \mathrm{MinMerge}(\mathscr{T}_{s-2}, \mathscr{T}_{s-1})$ will become a path of length $l-1$ and all labels in the path are 0, then for $k = s-1$ in Step 5,

$$\mathrm{Irr}(\mathscr{T}_{s-1}, l-1) \otimes d_s = (\underbrace{0, \ldots, 0}_{l-1}) \otimes d_s = (\underbrace{0, \ldots, 0}_{l-1}, d_s) = (\underbrace{0, \ldots, 0}_{l-1}, d_{s-1}) =: \alpha.$$

Since S contains at least one exponent vector β such that

$$\beta \geq (a_1, \ldots, a_{l-1}, d_{s-1}) \in \mathrm{Irr}(\mathscr{T}_{s-2}, l-1) \otimes d_{s-1},$$

we have α will be removed in the step $\mathrm{MaxMerge}(S, \alpha)$.

6. Scarf Complex and Enumeration Algorithm

Enumeration Algorithm

Input: a monomial ideal I in $\mathbb{K}[x_1, \ldots, x_n]$

Output: a set representing $\mathrm{Irr}(I)$

Step 1. Assume $\mathrm{Min}(I) = \{X^{\alpha_1}, \ldots, X^{\alpha_p}, x_1^{c_1}, \ldots, x_n^{c_n}\}$, where $c_i = \infty$ if I doesn't contain the power of x_i, $1 \leq i \leq n$.

Step 2. Let A be the exponent matrix of $\mathrm{Min}(I)$ such that the ith row is the exponent vector of X^{α_i}, $1 \leq i \leq p$ and the $(j+p)^{\text{th}}$ row is the exponent vector of $x_j^{c_j}$ for $1 \leq j \leq n$.

Step 3. Create a matrix B such that the rows of B are the first p rows of A.
Perform the ordinal deformation on each column of B, i.e., make each element in each column of B be unique and take value in $\{1, \ldots, p\}$. Let $\beta_1, \ldots, \beta_p$ be the row vector of B. Then $\{X^{\beta_1}, \ldots, X^{\beta_p}\}$ is an ordinal deformation of $\{X^{\alpha_1}, \ldots, X^{\alpha_p}\}$.

Step 4. Create a matrix C such that the first p rows of C are the rows of B and the $(j+p)^{\text{th}}$ row of C is the exponent vector of x_j^{p+1}, $1 \leq j \leq 2$.

Step 5. Choose a row e of B such that the first element of that row is 1. Let f_i be the exponent vector of x_i^{p+1}, $2 \leq i \leq n$. Then $intialFacetVec := \{e, f_2, \ldots, f_n\}$ is the (vector) set of exponent vectors of the initial facet $\{m_1 := X^e, x_2^{p+1}, \ldots, x_n^{p+1}\}$.

Step 6. Let $H : \mathbb{N}^n \to \mathbb{N}^n$ be a (hash) map. Set

$$H\{\mathrm{lcm}(initialFacet)\} = H\{(1, \underbrace{p+1, \ldots, p+1}_{n-1})\} = (1, \underbrace{0, \ldots, 0}_{n-1}).$$

Step 7. Let Q be a queue and push $initialFacetVec$ into Q.
While Q is NOT empty:
Pop an element $facetVec$ from Q.
Let $H_1 : \mathbb{N}^n \to \mathbb{N}$ be another (hash) map.
For i from 0 to $n-1$ do:
$H_1\{facetVect[i]\} = 0$.
For i from 0 to $n-1$ do:
If $H\{\mathrm{lcm}(facetVec)\}[i] = 1$, then
continue.
Then we try to find a substitute monomial of $X^{facetVec[i]}$ with respect to the facet $(X^{facetVec[0]}, \ldots, X^{facetVec[n-1]})$.)
Let $mPos$ be the index such that the $mPos^{\text{th}}$ vector of $facetVec$ has the largest degree in x_i among the vector set $facetVec \setminus \{\text{the } i^{\text{th}} \text{ vector of } facetVec\}$.
Let $index = -1$.
Let $temp = p + n + 2$.

For k from 0 to $p+n-1$ do:
  Let $index_1 = -2$.
  If the row vector $C[k]$ exists in the keys of H_1, then
    continue.
  For l from 0 to $n-1$ do:
    If $l \neq mPos$ and $l \neq i$ and $A[k][l] \geq \mathrm{lcm}(facetVec)[l]$, then
      let $index_1 = -1$,
      break.
    If $l = i$ and $A[k][l] \geq facetVec[mPos][i]$, then
      let $index_1 = -1$,
      break.
    If $l = mPos$ and $A[k][l] \leq facetVec[mPos][mPos]$, then
      let $index_1 = -1$,
      break.
  If $index_1 = -2$ and $A[k][mPos] < temp$, then
    let $index = k$,
    let $temp = A[k][mPos]$.
If $index = -1$, then
  continue.
Then we find a substitute monomial $X^{A[index]}$.
Let $substituteVec = A[index]$.
Create a new facet vector $facetVecCopy$, which has the same values as $facetVec$.
Let $facetVecCopy[i] = facet[mPos]$.
Let $facetVecCopy[mPos] = substituteVec$.
Then the maximum of the jth column of $facetVecCopy$ will be the jth element, i.e.,

$$\mathrm{lcm}(facetVecCopy)[j] = facetVecCopy[j][j],\ 1 \leq j \leq n.$$

If $\mathrm{lcm}(facetVecCopy)$ doesn't exist in the keys of H, then
  let $a = (\underbrace{0, \ldots, 0}_{n})$,
  let $a[mPos] = 1$,
  let $H\{\mathrm{lcm}(facetVecCopy)\} = a$,
  push $facetVecCopy$ into Q.
Let key be the keys of H.
Let H_2 be another hash map.
Let $\mathscr{C}$ consist of n hash maps.
For j from 0 to $n-1$ do:
  For i from 0 to $p+n-1$ do:
    $\mathscr{C}[j]\{C[i][j]\} = A[i][j]$.
For i from 0 to $\mathrm{length}(key)-1$ do:
  Let S be a list.
  Let $index = -2$.
  For j from 0 to $n-1$ do:
    If $\mathscr{C}[j]\{key[i][j]\} = 0$, then
      let $index = -1$,
      break.

Append $\mathscr{C}[j]\{key[i][j]\}$ to S.
If $index = -2$, then
let $H_2\{S\} = 0$.
Let $exponentVectors$ be the keys of H_2.
For $\alpha, \beta \in exponentVectors$, if $\alpha \leq \beta$, then remove α from $exponentVectors$.
Then $exponentVectors$ is the set of exponent vectors of the irreducible components.

7. Simulations

The monomial exponents are random numbers between 0 and $m-1$, where m is the maximal (signed) integer in C++.

(a) Irr algorithm.

# vars.	# min. generators	irred. components	run time
7	64	4673	165s
7	72	5501	321s
8	86		>20min.

(b) Enumeration algorithm

# vars.	# min. generators	irred. components	run time
7	73	7155	48s
7	67	4874	30s
8	78	22087	230s
8	85	23083	263s
9	84	54279	1009s

8. Upper Bound Problem

Set $\mathbb{N}_0 = \mathbb{N} \sqcup \{0\}$. Assume $\mathbb{K}$ is a field and I a monomial ideal in $\mathbb{K}[x_1, \ldots, x_n]$ which is minimally generated by the monomial set $\{m_1 := X^{\alpha_1}, \ldots, m_p := X^{\alpha_p}\}$, where each $\alpha_i = (a_{i,1}, \ldots, a_{i,n}) \in \mathbb{N}_0^n$ and $X^{\alpha_i} = x_1^{a_{i,1}} \cdots x_d^{a_{i,n}}$.

For a vector $\alpha = (a_1, \ldots, a_n) \in \mathbb{N}_0^n$, define

$$\alpha - 1 = (\max\{0, a_1 - 1\}, \ldots, \max\{0, a_n - 1\}).$$

Define $\Delta(I) \subseteq \mathbb{N}_0^n$ by

$$\begin{aligned}\Delta(I) &:= \{\alpha \in \mathbb{N}_0^n : X^\alpha \notin I\} \sqcup \{\alpha \in \mathbb{N}_0^n : X^\alpha \in I \text{ but } X^{\alpha-1} \notin I\} \\ &= \{\alpha \in \mathbb{N}_0^n \mid X^{\alpha-1} \notin I\}.\end{aligned}$$

For $m \in I$, we view I as an ideal $\mathbb{K}[x_1, \ldots, x_{n-1}][x_n]$, define $m|_{x_n} = \operatorname{coeff}_{x_n} m$ and define

$$I|_{x_n} = (m|_{x_n} : m \in I)\mathbb{K}[x_1, \ldots, x_{n-1}].$$

Then

$$I|_{x_n} = (m_1|_{x_n}, \ldots, m_p|_{x_n})\mathbb{K}[x_1, \ldots, x_{n-1}].$$

Assume I is generic and its minimal generating set $M = \{m_1, \ldots, m_p\}$ doesn't contain power of x_i, $1 \leq i \leq n$. Now let $\overline{M} = M \sqcup \{m_{p+1}\}$, where $m_{p+1} = x_1^{b_1} \ldots x_n^{b_n}$ satisfies these conditions:

(a) $b_n > \deg_{x_n} m_i$ for all $1 \leq i \leq p$,
(b) $\overline{M}$ is a minimal generating set for $\overline{I} := \langle \overline{M} \rangle$, and
(c) $\overline{I}$ is still generic.

Suppose all the distinct degree of x_n in $M \sqcup \{x_1^\infty, \dots, x_n^\infty\}$ are

$$0 = d_0 < d_1 < \cdots < d_{s-1} < d_s = \infty.$$

For $0 \le k \le s-1$, let

$$I_k = \langle \mathrm{coeff}_{x_n}(m) : m \in M \text{ and } \deg_{x_n} m \le d_k \rangle \subseteq \mathbb{K}[x_1, \dots, x_{n-1}],$$

and let $I_s = \mathbb{K}[x_1, \dots, x_{n-1}]$. Then $I_{s-1} = I|_{x_n}$.

Theorem 8.1. $\left|\mathrm{Irr}(\bar{I})\right| = |\mathrm{Irr}(I)| + \left|\mathrm{Irr}(\bar{I}|_{x_n}) \smallsetminus \mathrm{Irr}(I|_{x_n})\right|$.

Proof. Since $b_n > \deg_{x_n} m_i$ for all $1 \le i \le p$ and $\overline{M}$ is a minimal generating set for $\bar{I}$, we have $m_{p+1}|_{x_n} \notin I|_{x_n} = \langle m_1|_{x_n}, \dots, m_p|_{x_n} \rangle$. So

$$I|_{x_n} \subsetneq \bar{I}|_{x_n} = I_{x_n} + \langle m_{p+1}|_{x_n} \rangle.$$

Note that

$$\begin{aligned}
\mathrm{Irr}(I) &= \bigsqcup_{k=1}^{s} \big(\mathrm{Irr}(I_{k-1}) \smallsetminus \mathrm{Irr}(I_k)\big) \otimes d_k \\
&= \left[\bigsqcup_{k=1}^{s-1} \big(\mathrm{Irr}(I_{k-1}) \smallsetminus \mathrm{Irr}(I_k)\big) \otimes d_k\right] \sqcup \left[\big(\mathrm{Irr}(I_{s-1}) \smallsetminus \mathrm{Irr}(I_s)\big) \otimes \infty\right] \\
&= \left[\bigsqcup_{k=1}^{s-1} \big(\mathrm{Irr}(I_{k-1}) \smallsetminus \mathrm{Irr}(I_k)\big) \otimes d_k\right] \sqcup \left[\mathrm{Irr}(I_{s-1}) \otimes \infty\right] \\
&= \left[\bigsqcup_{k=1}^{s-1} \big(\mathrm{Irr}(I_{k-1}) \smallsetminus \mathrm{Irr}(I_k)\big) \otimes d_k\right] \sqcup \left[\mathrm{Irr}(I|_{x_n}) \otimes \infty\right].
\end{aligned}$$

So

$$|\mathrm{Irr}(I)| = \sum_{k=1}^{s-1} |\mathrm{Irr}(I_{k-1}) \smallsetminus \mathrm{Irr}(I_k)| + |\mathrm{Irr}(I|_{x_n})|.$$

Note that all the distinct degree of x_n in $\overline{M} \sqcup \{x_1^\infty, \dots, x_n^\infty\}$ are

$$0 = \bar{d}_0 = d_0 < \bar{d}_1 = d_1 < \cdots < \bar{d}_{s-1} = d_{s-1} < \bar{d}_s = b_n < \bar{d}_{s+1} = \infty.$$

For $0 \le k \le s$, let

$$\bar{I}_k = \langle \mathrm{coeff}_{x_n}(m) : m \in \overline{M} \text{ and } \deg_{x_n} m \le \bar{d}_k \rangle \subseteq \mathbb{K}[x_1, \dots, x_{n-1}],$$

and let $\bar{I}_{s+1} = \mathbb{K}[x_1, \dots, x_{n-1}]$. Then $\bar{I}_s = \bar{I}|_{x_n}$. Also, $\bar{I}_k = I_k$ for all $1 \le k \le s-1$. So

$$\begin{aligned}
\mathrm{Irr}(\bar{I}) &= \bigsqcup_{k=1}^{s+1} \big(\mathrm{Irr}(\bar{I}_{k-1}) \smallsetminus \mathrm{Irr}(\bar{I}_k)\big) \otimes \bar{d}_k \\
&= \bigsqcup_{k=1}^{s-1} \left[\big(\mathrm{Irr}(I_{k-1}) \smallsetminus \mathrm{Irr}(I_k)\big) \otimes d_k\right] \sqcup \left[\big(\mathrm{Irr}(I_{s-1}) \smallsetminus \mathrm{Irr}(\bar{I}_s)\big) \otimes b_n\right] \sqcup \left[\big(\mathrm{Irr}(\bar{I}_s) \smallsetminus \mathrm{Irr}(\bar{I}_{s+1})\big) \otimes \infty\right] \\
&= \bigsqcup_{k=1}^{s-1} \left[\big(\mathrm{Irr}(I_{k-1}) \smallsetminus \mathrm{Irr}(I_k)\big) \otimes d_k\right] \sqcup \left[\big(\mathrm{Irr}(I|_{x_n}) \smallsetminus \mathrm{Irr}(\bar{I}|_{x_n})\big) \otimes b_n\right] \sqcup \left[\mathrm{Irr}(\bar{I}|_{x_n}) \otimes \infty\right]
\end{aligned}$$

So

$$\begin{aligned}\big|\mathrm{Irr}(\bar{I})\big| &= \sum_{k=1}^{s-1}|\mathrm{Irr}(I_{k-1})\smallsetminus\mathrm{Irr}(I_k)| + \big|\mathrm{Irr}(I|_{x_n})\smallsetminus\mathrm{Irr}(\bar{I}|_{x_n})\big| + \big|\mathrm{Irr}(\bar{I}|_{x_n})\big| \\ &= |\mathrm{Irr}(I)| + \big|\mathrm{Irr}(I|_{x_n})\smallsetminus\mathrm{Irr}(\bar{I}|_{x_n})\big| + \big|\mathrm{Irr}(\bar{I}|_{x_n})\big| - |\mathrm{Irr}(I|_{x_n})| \\ &= |\mathrm{Irr}(I)| + |\mathrm{Irr}(I|_{x_n})| - \big|\mathrm{Irr}(I|_{x_n})\cap\mathrm{Irr}(\bar{I}|_{x_n})\big| + \big|\mathrm{Irr}(\bar{I}|_{x_n})\big| - |\mathrm{Irr}(I|_{x_n})| \\ &= |\mathrm{Irr}(I)| + \big|\mathrm{Irr}(\bar{I}|_{x_n})\big| - \big|\mathrm{Irr}(I|_{x_n})\cap\mathrm{Irr}(\bar{I}|_{x_n})\big| \\ &= |\mathrm{Irr}(I)| + \big|\mathrm{Irr}(\bar{I}|_{x_n})\smallsetminus\mathrm{Irr}(I|_{x_n})\big|. \end{aligned}$$

□

Let $A = \mathrm{Irr}(\bar{I}|_{x_n})\smallsetminus\mathrm{Irr}(I|_{x_n}) \subseteq \mathbb{K}[x_1,\ldots,x_{n-1}]$. When adding the monomial $m_{p+1}|_{x_n} = x_1^{b_1}\cdots x_n^{b_{n-1}}$ to $I|_{x_n}$, some irreducible components are destroyed, and some new irreducible components are generated. Then A is exactly the set of all new generated ones. Let

$$C_i = \{c\in A : c_i = b_i\},\ 1\le i\le n-1.$$

Then $\{C_1,\ldots,C_{n-1}\}$ forms a partition of A, i.e., $A = \sqcup_{i=1}^{n-1} C_i$.

For $i = 1,\ldots,n-1$, define

$$\begin{aligned}\bar{I}_{\le,b_i,x_n} &= \langle m_j|_{x_i,x_n} := (m_j|_{x_n})|_{x_i} : m_j\in\overline{M} \text{ and } \deg_{x_i} m_j \le b_i\rangle \\ &\subseteq \mathbb{K}[x_1,\ldots,x_{i-1},x_{i+1},\ldots,x_{n-1}],\end{aligned}$$

and

$$\begin{aligned}\bar{I}_{<,b_i,x_n} &= \langle m_j|_{x_i,x_n} := (m_j|_{x_n})|_{x_i} : m_j\in\overline{M} \text{ and } \deg_{x_i} m_j < b_i\rangle \\ &\subseteq \mathbb{K}[x_1,\ldots,x_{i-1},x_{i+1},\ldots,x_{n-1}].\end{aligned}$$

For $\nu\in\mathbb{N}_0^{n-1}$ and $i = 1,\ldots,n-1$, define

$$\nu|_i = (\nu_1,\ldots,\nu_{i-1},\nu_{i+1},\ldots,\nu_{n-1})\in\mathbb{N}_0^{n-2},$$

$$b_i * \nu = (\nu_1,\ldots,\nu_{i-1},b_i,\nu_{i+1},\ldots,\nu_{n-1})\in\mathbb{N}_0^{n-1}.$$

Mimic Theorem 5.1, we have for $\nu\in\mathrm{Irr}(\bar{I}|_{x_n})$ such that $\nu_i = b_i$, i.e., $b_i*\nu\in\mathrm{Irr}(\bar{I}|_{x_n})$, we have

$$\nu|_i\in\mathrm{Irr}(\bar{I}_{<,b_i,x_n})\smallsetminus\mathrm{Irr}(\bar{I}_{\le,b_i,x_n}),\ 1\le i\le n-1.$$

Since $\bar{I}$ is generic, there is no $m\in I|_{x_n}$ such that $\deg_{x_i} m = b_i$, $1\le i\le n-1$. So there is no $b_i*\nu\in\mathbb{N}^{n-1}$ such that $b_i*\nu\in\mathrm{Irr}(I|_{x_n})$, $1\le i\le n-1$. Hence for $i = 1,\ldots,n-1$, we have

$$\begin{aligned}C_i &= \{c\in A : c_i = b_i\} \\ &= \{b_i*\nu\in\mathbb{N}^{n-1} : b_i*\nu\in A\} \\ &= \{b_i*\nu\in\mathbb{N}^{n-1} : b_i*\nu\in\mathrm{Irr}(\bar{I}|_{x_n})\smallsetminus\mathrm{Irr}(I|_{x_n})\} \\ &= \{b_i*\nu\in\mathbb{N}^{n-1} : b_i*\nu\in\mathrm{Irr}(\bar{I}|_{x_n})\} \\ &= \{b_i*\nu\in\mathbb{N}^{n-1} : \nu|_i\in\mathrm{Irr}(\bar{I}_{<,b_i,x_n})\smallsetminus\mathrm{Irr}(\bar{I}_{\le,b_i,x_n})\} \\ &= \{\nu|_i\in\mathbb{N}^{n-2} : \nu|_i\in\mathrm{Irr}(\bar{I}_{<,b_i,x_n})\smallsetminus\mathrm{Irr}(\bar{I}_{\le,b_i,x_n})\}.\end{aligned}$$

When we delete $m_{p+1}|_{x_i,x_n}$ from the ideal $\bar{I}_{\le,b_i,x_n}$, any new irreducible component $\nu|_i$ that is not in $\mathrm{Irr}(\bar{I}_{\le,b_i,x_n})$ should satisfy $\nu_k\ge b_k$ for $k\in\{1,\ldots,n-1\}\smallsetminus\{i\}$. For $k\in\{1,\ldots,n-1\}\smallsetminus\{i\}$, let

$$\bar{I}_{<b_k,<,b_i,x_n} = \langle m_j|_{x_i,x_n} : m_j\in\overline{M} \text{ and } \deg_{x_i} m_j < b_i \text{ and } \deg_{x_k} m_j < b_k\rangle,$$

where x_k is not deleted from the generators in M or $\bar{I}_{<,b_i,x_n}$ if $k \neq i$, while x_i, x_n are deleted. Then (**conjecture 1**)

$$\begin{aligned}|C_i| &\leq \left|\mathrm{Irr}(\bar{I}_{<,b_i,x_n})\right| - \min_{1\leq k\leq n-1, k\neq i}\left\{\left|\mathrm{Irr}(\bar{I}_{<b_k,<,b_i,x_n})\right|\right\}\\ &= \left|\mathrm{Irr}(\bar{I}_{<,b_i,x_n})\right| - \left|\mathrm{Irr}(\bar{I}_{<b_{l_i},<,b_i,x_n})\right|,\end{aligned}$$

where $l_i \in \{1,\ldots,n-1\}\setminus\{i\}$ is such that $\left|\mathrm{Irr}(\bar{I}_{<b_{l_i},<,b_i,x_n})\right| = \min_{1\leq k\leq n-1,k\neq i}\left\{\left|\mathrm{Irr}(\bar{I}_{<b_k,<,b_i,x_n})\right|\right\}$. Given an ideal J, define

$$\mathrm{rank}_{b_i}(J) = \left|\{m \in \mathrm{Min}(J) : \deg_{x_i} m < b_i\}\right|,\ 1 \leq i \leq n-1.$$

Let $p_i = \mathrm{rank}_{b_i}(I)$ for $i = 1,\ldots,n-1$ and assume $p_1 = \max\{p_1,\ldots,p_{n-1}\}$.

Let $\rho_i := \left|\mathrm{Min}(\bar{I}_{<b_{l_i},<,b_i,x_n})\right|$. Since

$$|C_i| + \left|\mathrm{Irr}(\bar{I}_{<b_{l_i},<,b_i,x_n})\right| \leq \left|\mathrm{Irr}(\bar{I}_{<,b_i,x_n})\right|$$

holds for all possible $\bar{I}_{<b_{l_i},<,b_i,x_n}$, we have

$$|C_i| + C_{n-2}(\rho_i) \leq C_{n-2}(p_i),\ 1 \leq i \leq n-1.$$

Since

$$\bar{I}_{<,b_i,x_n} = \bar{I}_{<b_{l_i},<,b_i,x_n} + \langle m_j|_{x_i,x_n} : m_j \in M \text{ and } \deg_{x_i} m_j < b_i \text{ and } \deg_{x_{l_i}} m_j \geq b_{l_i}\rangle,$$

and

$$\begin{aligned}&\left|\{m_j|_{x_i,x_n} : m_j \in M \text{ and } \deg_{x_i} m_j < b_i \text{ and } \deg_{x_{l_i}} m_j \geq b_{l_i}\}\right|\\ \leq &\left|\{m_j|_{x_i,x_n} : m_j \in M \text{ and } \deg_{x_{l_i}} m_j \geq b_{l_i}\}\right|\\ = &\, p - p_{l_i},\end{aligned}$$

we have

$$\rho_i \geq p_i - (p - p_{l_i}),\ 1 \leq i \leq n-1.$$

So

$$|C_i| \leq C_{n-2}(p_i) - C_{n-2}(\rho_i) \leq C_{n-2}(p_i) - C_{n-2}(p_i - (p - p_{l_i})),\ 1 \leq i \leq n-1.$$

In **??**, it is proved that $C_{n-2}(p) = \Theta(p^{[\frac{n}{2}]-1})$. Since $C_{n-2}(p_i) - C_{n-2}(p_i - (p - p_{l_i}))$ is super linear on p_i, its value increases when p_i increases. Hence (**conjecture 2**)

$$|C_i| \leq C_{n-2}(p_1) - C_{n-2}(2p_1 - p).$$

Thus,

$$\begin{aligned}C_n(p+1) - C_n(p) &\leq \sum_{i=1}^{n-1}|C_i|\\ &\leq (n-1)C_{n-2}(p_1) - (n-1)C_{n-2}(2p_1 - p)\\ &< (n-1)C_{n-2}(p_1) - (n-2)C_{n-2}(2p_1 - p).\end{aligned}$$

We assume $C_{n-2}(p) = f(n,p)p^{[\frac{n}{2}]-1}$, where $f(n,p)$ is a function on n and p, and $f(n,p)$ decreases when p increases. Since $2p_1 - p \leq p_1$, we have $C_{n-2}(2p_1 - p) \geq f(n,p)(2p_1 - p)^{[\frac{n}{2}]-1}$. Let $q = [\frac{n}{2}] - 1$. Then

$$C_n(p+1) - C_n(p) < f(n)\big((n-1)p_1^q - (n-2)(2p_1 - p)^q\big).$$

Let $y = \frac{p_1}{p}$. Since $0 < y < 1$, we have $g(y) = (n-1)y^q - (n-2)(2y-1)^q$ has a maximal value $\frac{n-1}{2}y^{q-1}$ when $g'(y) = 0$. So $(n-1)y^{q-1} = 2(n-2)(2y-1)^{q-1}$ and $\frac{n-1}{n+1} < y < \frac{n}{n+1}$. Hence

$$C_n(p+1) - C_n(p) < f(n)p^q \frac{n-1}{2} y^{q-1} \leq \frac{(n-1)}{2y} C_{n-2}(p) < \frac{n+1}{2} C_{n-2}(p).$$

In **??**, it is shown that $C_3(p) = 2p+1$ and $C_4(p) \leq (p^2+5p+2)/2$. Therefore, for $n > 2$,

$$C_n(p) \leq \begin{cases} \frac{n+1}{2} p^{\frac{n-1}{2}} & \text{if } n \text{ is odd,} \\ \frac{n+2}{12} p^{\frac{n}{2}} & \text{if } n \text{ is even.} \end{cases}$$

Email address: `sgao@ces.clemson.edu`

School of Mathematical and Statistical Sciences, Clemson University, O-301 Martin Hall, Box 340975, Clemson, SC 29634, USA

Email address: `sgao@ces.clemson.edu`

School of Mathematical and Statistical Sciences, Clemson University, O-301 Martin Hall, Box 340975, Clemson, SC 29634, USA

Email address: `wei6@clemson.edu`

School of Mathematical and Statistical Sciences, Clemson University, E-008 Martin Hall, Box 340975, Clemson, SC 29634, USA